

Prediction of Acute vs Chronic workload ratio for players

Client: Departamento de Data del Club

Company: *[Team name]* – Group A

Date: March 2026

1. Context

The Club's Data Department is the internal advanced analytics unit that supports the physical-training and performance departments. Our mission is to transform workload data and performance metrics into actionable information that the technical staff and fitness coaches can use to optimize athlete performance and reduce injury risk.

Over the last two seasons we have heavily invested in capturing workload data and performance metrics for every training session and match. While the data are collected and stored correctly, our technical staff and fitness coaches have repeatedly expressed the need for intuitive, interactive tools that translate this data into useful decision-making information without writing code or querying databases. Specifically, they want a tool that lets them predict and visualize the Acute vs Chronic Workload Ratio (ACWR) of athletes based on the proposed training type, enabling informed decisions to optimize performance and reduce injury risk.

The goal of this project is to design, build, and deliver a prediction and visualization tool for the Acute vs Chronic Workload Ratio (ACWR) for football players. The ACWR is a key indicator for assessing injury risk and optimizing performance.

2. Problem statement

Our analysts and fitness coaches need to answer questions such as:

- What will the ACWR of our athletes be if we run a high-intensity training tomorrow?
- Which athletes are most likely to have a high ACWR based on their training and injury history?

Today the technical team does not have this information and plans training based on their own experience and recent metric values.

We need a comprehensive solution—from raw workload and performance metrics to interactive ACWR visualization—that the technical staff and fitness coaches can manage independently, allowing them to make informed decisions and optimize athlete performance more efficiently.

3. Scope

Students are asked to develop a complete solution that predicts the ACWR of football athletes based on the proposed training type. The solution must include:

3.1 Data

Input data will include:

- Training and match history of athletes (intensity, duration, exercise type, etc.)
- Workload and fatigue metrics

Data will be available in CSV or JSON format and will be provided at the start of the project.

3.2 Data Pipeline

1. **Data Preparation** – Clean, transform, and prepare data for analysis.
2. **Predictive Model Development** – Build a machine-learning model that predicts ACWR from training characteristics and athlete metrics. Explore different algorithms and modeling techniques.
3. **Model Validation** – Evaluate model performance using appropriate metrics (e.g., RMSE, MAE, R^2).

3.3 Visualization

Develop an interactive application that lets the technical team:

- Select the proposed training type (e.g., strength training, cardiovascular training, etc.) for the coming days.
- Visualize the predicted ACWR for the next 15 days for each athlete based on the proposed training.

The application should be user-friendly and provide clear, understandable visualizations.

3.4 Key References

1. **Pipeline and Model Code** – Databricks notebooks and/or repositories containing code for data preparation and predictive model development.
2. **Interactive Application** – A deployable app that allows interaction with the predictive model.
3. **Technical Documentation** – A document describing the solution architecture, data model used, methodology followed, and a user guide for the interactive app.
4. **Final Presentation** – A 15–20 minute executive-level PowerPoint presentation accompanied by a live demo of the app.

4. Referencias Clave

ACWR definition (<https://www.scienceforsport.com/acutechronic-workload-ratio>).

4.1 Expert Mentoring Sessions

Each team will have access to a maximum of 4 hours of remote tutoring with a Club Data Department expert during the project. Teams should propose a tutoring schedule as part of their Week 2 work plan.

Suggested distribution (not mandatory):

Session	Suggested Timing	Suggested Duration	Focus
1	End of Week 2	45 min	Validate approach and data understanding
2	End of Week 4	60 min	Review pipeline and model validation
3	End of Week 6	60 min	Feedback on visualization and what-if design
4	End of Week 7	45 min	Pre-delivery review and presentation rehearsal

5. Work Plan Submission (Week 2)

By the end of Week 2 each team must submit a document containing:

1. **Team Identity** – Team name and a brief description of the team’s approach.
 2. **Roles and Responsibilities** – Assigned role for each team member (e.g., Project Lead, Data Engineer, Visualization Specialist, ML/Analytics Lead) with a clear description of responsibilities.
 3. **8-Week Activity Plan** – A Gantt diagram or equivalent breakdown of tasks by role and week.
 4. **Mentoring Schedule** – Proposed dates, durations, and agenda items for each of the (up to) four hours mentoring sessions.
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6. Licensing

All code and deliverables must be submitted under an open-source license, such as the Apache License, Version 2.0.

Club Data Department